ENGINEERING CHEMISTRY · UNIT II

Electrochemistry & Applications

Complete Formula & Concept Revision Sheet

Last-day revision toolkit — all key formulas, reactions, mechanisms, and exam tips in one place

PROGRAM	SEMESTER	SUBJECT	UNIT
B.Tech 1st Year	Semester I	Engg. Chemistry	Unit II

ELECTROCHEMICAL CELLS & ELECTRODES

Electrochemical Cell

- Device that converts **chemical energy → electrical energy**
- Consists of two half-cells (electrodes) connected by a salt bridge
- Involves spontaneous redox reactions

Electrolytic Cell

- Converts **electrical energy → chemical energy**
- Non-spontaneous reactions driven by external EMF
- Used in electroplating, electrolysis

Feature	Galvanic (Electrochemical)	Electrolytic Cell
Energy Conversion	Chemical → Electrical	Electrical → Chemical
Reaction Type	Spontaneous ($\Delta G < 0$)	Non-spontaneous ($\Delta G > 0$)
Anode	Negative (−)	Positive (+)
Cathode	Positive (+)	Negative (−)
External Power	Not required	Required
Example	Daniel Cell, Batteries	Electroplating, Electrolysis

CELL NOTATION & EMF

Cell Representation (Daniel Cell):

$Zn(s)\ |\ Zn^{2+}(aq)\ ||\ Cu^{2+}(aq)\ |\ Cu(s)$

Left = Anode (oxidation) | Double line = salt bridge | Right = Cathode (reduction)

EMF OF CELL

Also written as: $E_{cell} = E_{reduction}(+) - E_{reduction}(-)$

STANDARD EMF

Measured at 298K, 1 atm, 1M concentration

RELATIONSHIP WITH GIBBS ENERGY

n = moles of electrons transferred; F = 96485 C/mol (Faraday's constant)

NERNST EQUATION

GENERAL (THERMODYNAMIC) FORM

$R = 8.314$ J/mol·K | T = Temperature (K) | Q = Reaction quotient

LOGARITHMIC FORM (AT 298 K)

$0.0592 = (2.303 \times R \times 298) / F \approx 0.0592$ V (use 0.059 in MCQs)

REACTION QUOTIENT Q (FOR AA + BB → CC + DD)

At equilibrium: $Q = K_{eq}$ and $E_{cell} = 0$

✓ Key Points:

- $E^\circ_{cell} > 0$ → spontaneous reaction
- $E^\circ_{cell} < 0$ → non-spontaneous
- Higher concentration at anode → lower EMF

⚠ Common Mistakes:

- Using log instead of ln (×2.303)
- Wrong sign: $E_{anode} - E_{cathode}$
- Forgetting to substitute T in Kelvin

PRIMARY & SECONDARY CELLS

PRIMARY CELL — ZINC-AIR BATTERY

Construction

- **Anode:** Zinc (Zn) powder
- **Cathode:** Porous carbon + air (O_2)
- **Electrolyte:** KOH (alkaline)
- **Separator:** Ion-exchange membrane

Key Facts

- High energy density (~1350 Wh/kg)
- Voltage: ~1.65 V
- Non-rechargeable (primary)
- Uses atmospheric oxygen as cathode reactant

Anode (oxidation)

$Zn \rightarrow Zn^{2+} + 2e^-$ OR $Zn + 2OH^- \rightarrow Zn(OH)_2 + 2e^-$

Cathode (reduction)

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

Overall Reaction

$2Zn + O_2 \rightarrow 2ZnO$

SECONDARY CELL — NICKEL-CADMIUM (NI-CD)

Construction

- **Anode:** Cadmium (Cd)
- **Cathode:** Nickel oxyhydroxide [NiO(OH)]
- **Electrolyte:** KOH (30%)
- Voltage: ~1.2 V per cell

Applications

- Portable electronics, power tools
- Emergency lighting
- Aircraft & satellite systems
- Rechargeable (500–1000 cycles)

Discharge — Anode (oxidation)

$Cd + 2OH^- \rightarrow Cd(OH)_2 + 2e^-$

Discharge — Cathode (reduction)

$2NiO(OH) + 2H_2O + 2e^- \rightarrow 2Ni(OH)_2 + 2OH^-$

Overall Discharge

$Cd + 2NiO(OH) + 2H_2O \rightarrow Cd(OH)_2 + 2Ni(OH)_2$

Charging (reverse of discharge)

$Cd(OH)_2 + 2Ni(OH)_2 \rightarrow Cd + 2NiO(OH) + 2H_2O$

SECONDARY CELL — LITHIUM-ION BATTERY

Construction

- **Anode:** Graphite (intercalated Li)
- **Cathode:** $LiCoO_2$ / $LiMnO_2$
- **Electrolyte:** $LiPF_6$ in organic solvent
- Voltage: ~3.6 V per cell

Advantages

- High energy density (~150–200 Wh/kg)
- Low self-discharge (<5%/month)
- No memory effect
- Long cycle life (500–2000 cycles)

Discharge — Cathode

$$LiCoO_2 \rightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$$

Discharge — Anode

$$C_6 + xLi^+ + xe^- \rightarrow Li_xC_6$$

Overall

$$LiCoO_2 + C_6 \rightleftharpoons Li_{1-x}CoO_2 + Li_xC_6 \quad (\rightleftharpoons = \text{reversible, rechargeable})$$

📊 BATTERY COMPARISON TABLE

Feature	Zinc-Air	Ni-Cd	Li-Ion
Type	Primary	Secondary	Secondary
Voltage	1.65 V	1.2 V	3.6 V
Anode	Zinc	Cadmium	Graphite/Li
Cathode	O_2 (air)	NiO(OH)	$LiCoO_2$
Electrolyte	KOH	KOH	$LiPF_6$/organic
Energy Density	Very High	Moderate	High
Rechargeable?	No	Yes	Yes
Memory Effect	N/A	Yes	No
Applications	Hearing aids, watches	Power tools, aviation	Phones, EVs, laptops

FUEL CELLS

What is a Fuel Cell?

- Electrochemical device that **continuously converts** chemical energy (from fuel) to electrical energy
- Operates as long as fuel and oxidant are supplied
- No combustion — directly produces electricity
- Efficiency: 40–70% (vs ~35% for thermal plants)

Fuel Cell vs Battery

Feature	Fuel Cell	Battery
Fuel	External (continuous)	Internal (stored)
Recharging	Not needed	Needed
Reactant	H_2, O_2, hydrocarbons	Electrodes
Lifetime	Very long	Limited cycles

HYDROGEN-OXYGEN FUEL CELL

Construction

- **Anode:** Porous carbon + platinum catalyst
- **Cathode:** Porous carbon + platinum catalyst
- **Electrolyte:** KOH (alkaline type) or H_3PO_4 (acid type)
- Temperature: 60–200°C

Working Principle

- H_2 supplied at anode → oxidized
- O_2 supplied at cathode → reduced
- Ions travel through electrolyte
- Electrons flow through external circuit → current
- Byproduct: only water (clean!)

Anode (oxidation) — Alkaline

$$2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$$

Cathode (reduction) — Alkaline

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$$

Overall Reaction

$$2H_2 + O_2 \rightarrow 2H_2O + \text{Electrical Energy}$$

Anode (oxidation) — Acidic

$$2H_2 \rightarrow 4H^+ + 4e^-$$

Cathode (reduction) — Acidic

$$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$$

THEORETICAL EFFICIENCY OF H_2–O_2 FUEL CELL

✓ Advantages:

- Zero emission (only water produced)
- High efficiency compared to heat engines
- Quiet operation, no moving parts
- Continuous operation (no recharge)
- Scalable power output

Applications

- Space vehicles (NASA Apollo)
- Fuel cell electric vehicles (FCEVs)
- Submarines & military
- Portable power generation
- Backup power for hospitals

Definition

Corrosion is the **deterioration or degradation** of a metal or alloy due to its reaction with the environment (chemical or electrochemical), resulting in loss of material and properties.

Importance of Studying Corrosion

- Causes ~3–4% of GDP loss worldwide
- Structural failures in bridges, pipelines
- Safety hazards in nuclear plants
- Leads to development of protective coatings

ELECTROCHEMICAL THEORY OF CORROSION

Principle: Corrosion occurs due to formation of local galvanic cells on the metal surface where anodic and cathodic regions exist simultaneously.

Anodic Reaction (Metal Dissolution / Oxidation)

$$M \rightarrow M^{n+} + ne^-$$

Cathodic Reaction (in neutral/alkaline, aerated solution)

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$$

Cathodic Reaction (in acidic solution)

$$2H^+ + 2e^- \rightarrow H_2\uparrow$$

Rust Formation (Iron)

$$4Fe + 3O_2 + 2H_2O \rightarrow 2Fe_2O_3 \cdot H_2O \text{ (hydrated ferric oxide / rust)}$$

TYPES OF CORROSION

1. Differential Aeration Corrosion

- Caused by **unequal oxygen concentration** at different parts of metal
- Region with **less O_2** → becomes **anode** → corrodes
- Region with **more O_2** → becomes **cathode**

Examples:

- Pitting corrosion under debris
- Iron pipe buried in soil at different depths
- Waterline corrosion on ship hull

2. Galvanic Corrosion

- Occurs when **two dissimilar metals** are in electrical contact in an electrolyte
- **More active metal** (higher in galvanic series) → acts as **anode** → corrodes
- Less active metal → cathode → protected
- Rate increases with larger potential difference

Examples:

- Zinc–Copper couple in seawater (Zn corrodes)
- Steel rivets in copper sheet
- Iron pipe connected to brass fitting

3. Dry (Chemical) Corrosion

- Occurs in absence of moisture/electrolyte
- Direct chemical reaction between metal and gas (O_2, SO_2, H_2S, etc.)
- Common at **high temperatures**
- Forms metal oxide layer on surface

Metal Oxide Formation:

General

$M + O_2 \rightarrow M_xO_y$

Iron

$3Fe + 2O_2 \rightarrow Fe_3O_4$ (magnetite)

Aluminium

$4Al + 3O_2 \rightarrow 2Al_2O_3$ (protective)

Feature	Differential Aeration	Galvanic	Dry/Chemical
Cause	O_2 concentration difference	Dissimilar metals contact	Gas-metal reaction
Medium	Electrolyte required	Electrolyte required	No electrolyte needed
Temperature	Normal	Normal	Elevated temperature
Anode	Low O_2 region	Active metal	Whole metal surface
Example	Pitting, waterline	Zn-Cu, Fe-Cu joints	Fe_3O_4, Al_2O_3 formation

PILLING-BEDWORTH RATIO & CORROSION FACTORS

PILLING-BEDWORTH RATIO (PBR)

PBR FORMULA

M = Molar mass; ρ = Density; n = moles of metal in oxide formula

PBR < 1

- Oxide layer is **porous/cracked**
- Does NOT protect metal
- Corrosion continues
- Example: Alkali metals (Na, K, Ca)

PBR ≈ 1 (1–2)

- Oxide layer is **compact & adherent**
- Forms a **protective coating**
- Prevents further corrosion
- Example: Al_2O_3 (PBR = 1.28)

PBR > 2

- Oxide layer under **compressive stress**
- Flakes off / spalls
- Not protective
- Example: Fe_3O_4 (PBR = 2.1)

Metal	Oxide	PBR Value	Nature
Sodium (Na)	Na_2O	0.57	Non-protective
Magnesium (Mg)	MgO	0.81	Non-protective
Aluminium (Al)	Al_2O_3	1.28	Protective
Chromium (Cr)	Cr_2O_3	2.02	Protective
Iron (Fe)	Fe_3O_4	2.10	Borderline/Spalls
Tungsten (W)	WO_3	3.40	Non-protective

FACTORS AFFECTING CORROSION RATE

Nature of Metal

- **Position in EMF series:** More active (anodic) metals corrode faster
- **Purity:** Impurities create galvanic couples → accelerate corrosion
- **Physical state:** Stressed/strained regions corrode faster
- **Passivation:** Al, Cr, Ti form protective oxide layers

Temperature

- Higher temp → higher diffusion rate → faster corrosion
- Dissolved O_2 decreases with temperature (counteracts)
- Kinetic factor dominates at high temperatures

Environment & Moisture

- **Humidity:** Critical humidity (60–65%) required for corrosion onset
- **Electrolytes:** Presence of salt (NaCl) accelerates corrosion
- **pH:** Acidic conditions → faster corrosion
- **O_2 concentration:** Higher O_2 → faster cathodic reaction

Impurities & Alloy Composition

- Impurities act as cathodes → increase local current
- Grain boundaries are more susceptible
- Alloying with Cr, Ni improves corrosion resistance

CATHODIC PROTECTION

1. Sacrificial Anode Method

- A more **active metal** (Zn, Mg, Al) is connected to the metal to be protected
- Active metal acts as anode → sacrificially corrodes
- Protected metal acts as cathode → protected
- Anode replaced periodically

Sacrificial Anode Reaction

$Mg \rightarrow Mg^{2+} + 2e^-$ (Mg corrodes, Fe protected)

Applications:

- Underground pipelines
- Ship hulls (Zn blocks)
- Water heater tanks

2. Impressed Current Method

- External DC current applied to make protected metal a **cathode**
- Inert anode used (graphite, platinum)
- Current drives away corrosive reactions
- Requires continuous power supply

Applications:

- Long-distance gas/oil pipelines
- Offshore platforms
- Bridges, harbor structures
- Reinforced concrete structures

ANODIC PROTECTION

- External anodic current applied to promote formation of a **stable passive oxide film**
- Metal enters passive region → corrosion rate drops drastically
- Suitable only for **passive metals** (Fe, Ni, Cr, stainless steel)
- Used in chemical industry (sulfuric acid storage, fertilizer plants)

Feature	Cathodic (Sacrificial)	Cathodic (Impressed)	Anodic
Mechanism	Active metal sacrificed	External DC current	Passive film formation
Power Required	No	Yes	Yes
Anode Material	Zn, Mg, Al	Graphite, Pt	Protected metal
Maintenance	Anode replacement	Power monitoring	Current control
Applications	Ships, pipelines	Buried structures	Acid storage tanks

OTHER PREVENTION METHODS

Surface Coatings

- **Metallic coatings:** Galvanizing (Zn on Fe), Tinning, Chromium plating
- **Chemical conversion:** Anodizing, phosphating, chromating
- **Organic coatings:** Paints, lacquers, varnishes, enamels

Alloying & Design

- Stainless steel (Fe + 12% Cr) → passive oxide layer
- Avoid dissimilar metal contacts
- Use corrosion inhibitors in electrolyte
- Design to avoid water retention

ELECTROPLATING & ELECTROLESS PLATING

SECTION 07

ELECTROPLATING

Principle

- Uses **electrolytic cell** principle
- Object to be plated = **cathode**
- Metal to be deposited = **anode**
- Electrolyte = salt solution of plating metal
- External DC current drives deposition

Process Steps

- **Step 1:** Surface preparation (cleaning, pickling)
- **Step 2:** Strike plating (thin flash layer)
- **Step 3:** Main plating bath
- **Step 4:** Post-treatment (rinsing, drying)

Nickel Plating

Cathode (Object)

$Ni^{2+} + 2e^- \rightarrow Ni$ (deposited)

Anode (Ni metal)

$Ni \rightarrow Ni^{2+} + 2e^-$

Electrolyte: $NiSO_4 + NiCl_2 + H_3BO_3$ (Watts Bath) | **pH:** 3.5–4.5 | **Temp:** 45–65°C
Uses: Decorative, hardening surfaces, corrosion resistance for automotive parts

Copper Plating

Cathode

$Cu^{2+} + 2e^- \rightarrow Cu$ (deposited)

Anode (Copper metal)

$Cu \rightarrow Cu^{2+} + 2e^-$

Electrolyte: $CuSO_4 + H_2SO_4$ (acid bath) | **Uses:** PCBs, jewelry, base for other platings

FARADAY'S LAW OF ELECTROLYSIS

m = mass deposited (g) | M = molar mass | I = current (A) | t = time (s) | n = valency | F = 96485 C/mol

ELECTROLESS PLATING

Definition & Principle

- **No external electric current required**
- Deposition by **autocatalytic chemical reduction**
- Reducing agent in bath provides electrons
- Process is self-sustaining once initiated
- Common reducing agents: NaH_2PO_2 (hypophosphite), formaldehyde, hydrazine

Mechanism (Electroless Ni)

Nickel deposition

$$Ni^{2+} + 2e^- \rightarrow Ni^0$$

Reducing agent (hypophosphite)

$$H_2PO_2^- + H_2O \rightarrow H_2PO_3^- + 2H^+ + 2e^-$$

Overall

$$Ni^{2+} + H_2PO_2^- + H_2O \rightarrow Ni^0 + H_2PO_3^- + 2H^+$$

Feature	Electroplating	Electroless Plating
Power Supply	External DC required	Not required
Reduction Agent	Electrons (external)	Chemical reducing agent
Substrate	Must be conducting	Any surface (metals, plastics)
Uniformity	Non-uniform on complex shapes	Highly uniform thickness
Cost	Lower (simpler setup)	Higher (complex chemistry)
Applications	Jewelry, automotive, PCBs	PCBs, aerospace, plastic plating
Adhesion	Good	Excellent

Q **What is EMF of a cell?**

A Potential difference between the two electrodes; driving force for current flow in the circuit. Unit: Volt (V)

Q **Nernst equation is used for?**

A Calculating cell potential under non-standard conditions (variable concentration, temperature)

Q **What is the standard hydrogen electrode (SHE) potential?**

A $E° = 0.00$ V (by convention); used as reference for all standard electrode potentials

Q **PBR significance?**

A Indicates protective nature of metal oxide layer. PBR 1–2 = protective; <1 or >2 = non-protective

Q **What happens at anode in galvanic cell?**

A Oxidation occurs. Anode is the negative terminal. Metal loses electrons: $M \rightarrow M^{n+} + ne^-$

Q **What happens at cathode in galvanic cell?**

A Reduction occurs. Cathode is positive terminal. Ions gain electrons: $M^{n+} + ne^- \rightarrow M$

Q **Byproduct of H_2–O_2 fuel cell?**

A Only water (H_2O). Makes it zero-emission. Overall: $2H_2 + O_2 \rightarrow 2H_2O$

Q **What is galvanic corrosion?**

A Corrosion when two dissimilar metals contact in electrolyte; more active metal corrodes (becomes anode)

Q **Faraday's constant value?**

A F = 96485 C/mol ≈ 96500 C/mol. Charge of one mole of electrons

Q **What is differential aeration corrosion?**

A Corrosion due to unequal O_2 concentration; region with less O_2 becomes anode and corrodes

Q **Li-ion battery cathode material?**

A $LiCoO_2$ (lithium cobalt oxide). Anode: graphite. Electrolyte: $LiPF_6$ in organic solvent

Q **Why is Al corrosion resistant?**

A Al_2O_3 has PBR = 1.28 (protective range). Dense, adherent oxide layer prevents further attack

Q **Sacrificial anode method uses?**

A More active metals (Zn, Mg, Al) connected to protected metal; they corrode preferentially as anode

Q **What makes Ni-Cd battery rechargeable?**

A Reversible electrochemical reactions: $Cd/Cd(OH)_2$ and $Ni(OH)_2/NiO(OH)$ redox couple

Q **Electroless plating requires?**

A No external current; uses chemical reducing agent (NaH_2PO_2) for autocatalytic metal deposition

Q **ΔG and E cell relation?**

A $\Delta G° = -nFE°_{cell}$. Negative ΔG = spontaneous; positive E°cell = spontaneous reaction

Q **Salt bridge function?**

A Maintains electrical neutrality in both half-cells by allowing ion transfer while preventing direct mixing

Q **Dry corrosion mechanism?**

A Direct chemical reaction between metal and gas (O_2, SO_2, Cl_2) without moisture; common at high temperatures

Q **Zinc-air battery electrolyte?**

A KOH (potassium hydroxide) — alkaline electrolyte. Uses atmospheric O_2 as cathode reactant

Q **What is passivation?**

A Formation of thin, stable, adherent oxide film on metal surface that inhibits further corrosion (e.g., Al, Cr, stainless steel)

Q **Nernst equation constant at 298K?**

A 0.0592/n (using $\log_{10}$). Derived from (2.303RT/F) where R=8.314, T=298, F=96485

Q **Impressed current method uses?**

A External DC current to make protected structure cathodic; inert anode (graphite/Pt); used for buried pipelines

Q **Fuel cell vs battery key difference?**

A Fuel cell uses continuous external fuel supply; battery stores chemical energy internally and needs recharging

Q **Rust formula?**

A $Fe_2O_3 \cdot H_2O$ (hydrated ferric oxide). Formed by: $4Fe + 3O_2 + 2H_2O \rightarrow 2Fe_2O_3 \cdot H_2O$

Q **In electroplating, which electrode is the object?**

A Cathode (negative terminal). Metal ions deposit on it. Anode is the plating metal source

Q **Critical humidity for corrosion?**

A ~60–65% relative humidity. Below this, insufficient moisture for electrolyte film formation on metal surface

EXAM TIPS & COMMON MISTAKES

NERNST EQUATION MISTAKES

✗ **Wrong:** Using log without ×2.303 factor
✓ **Correct:** $E = E° - (0.0592/n) \log Q$ at 298K

✗ **Wrong:** $E_{cell} = E°_{anode} - E°_{cathode}$
✓ **Correct:** $E_{cell} = E°_{cathode} - E°_{anode}$

✗ **Wrong:** Putting T in °C in Nernst equation
✓ **Correct:** T must be in Kelvin (K = °C + 273)

✗ **Wrong:** Flipping Q (products/reactants reversed)
✓ **Correct:** Q = [Products] / [Reactants] (follow stoichiometry)

CELL & EMF MISTAKES

✗ **Wrong:** Anode = positive in galvanic cell
✓ **Correct:** Anode = negative (−) in galvanic cell; cathode = positive (+)

✗ **Wrong:** Anode = negative in electrolytic cell
✓ **Correct:** Anode = positive (+) in electrolytic cell

✗ **Wrong:** Omitting units for E (cell potential)
✓ **Correct:** Always express in Volts (V)

BATTERY & FUEL CELL MISTAKES

✗ **Wrong:** Confusing Ni-Cd charging/discharging reactions
✓ **Correct:** Discharge: Cd oxidizes; Charging: $Cd(OH)_2 \rightarrow Cd$ (reverse)

✗ **Wrong:** Saying fuel cells need recharging like batteries
✓ **Correct:** Fuel cells operate continuously as long as fuel is supplied

✗ **Wrong:** Fuel cell byproduct = CO_2
✓ **Correct:** H_2-O_2 fuel cell produces ONLY water (zero emission)

CORROSION MISTAKES

✗ **Wrong:** Higher O_2 region = anode (corrodes)
✓ **Correct:** Lower O_2 region = anode (corrodes) in differential aeration

✗ **Wrong:** PBR = 1.5 means non-protective
✓ **Correct:** PBR between 1 and 2 is protective (compact oxide)

✗ **Wrong:** Any PBR > 1 = protective
✓ **Correct:** PBR > 2 = non-protective (spalling due to stress)

✗ **Wrong:** Zinc coat on iron — Zn is cathode
✓ **Correct:** Zn is anode (more active); sacrificially protects Fe even when coating is scratched

SCORING TIPS

✓ High-Yield Topics:

- Nernst equation numerical problems (5 marks)
- Battery reaction equations (write balanced half-reactions)
- PBR calculation and interpretation (2–3 marks)
- Corrosion mechanism with diagram description
- Electroplating Faraday's law numericals

✓ Remember These Numbers:

- F = 96485 C/mol ≈ 96500 C/mol
- R = 8.314 J/mol·K
- 0.0592 = (2.303 × 8.314 × 298) / 96485
- Li-ion voltage: ~3.6 V | Ni-Cd: ~1.2 V | Zn-Air: ~1.65 V
- PBR of Al_2O_3 = 1.28 (protective example)

QUICK REVISION SHEET — ALL FORMULAS & KEY REACTIONS

CORE FORMULAS

EMF

GIBBS ENERGY

NERNST (GENERAL)

NERNST (AT 298K)

PBR

FARADAY'S LAW

KEY REACTIONS

Rust

$4Fe + 3O_2 + 2H_2O \rightarrow 2Fe_2O_3 \cdot H_2O$

H_2–O_2 Fuel Cell

$2H_2 + O_2 \rightarrow 2H_2O$

Zn–Air (Overall)

$2Zn + O_2 \rightarrow 2ZnO$

Ni–Cd Discharge

$Cd + 2NiO(OH) + 2H_2O \rightarrow Cd(OH)_2 + 2Ni(OH)_2$

Corrosion (Cathodic)

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

Corrosion (Anodic)

$M \rightarrow M^{n+} + ne^-$

Al Passivation

$4Al + 3O_2 \rightarrow 2Al_2O_3$

📌 KEY DEFINITIONS

- **EMF:** Potential difference driving current flow
- **Standard electrode potential:** Potential at 1M, 298K, 1 atm vs SHE
- **Corrosion:** Metal degradation by environment (electrochemical/chemical)
- **PBR:** Ratio of oxide volume to metal volume consumed
- **Passivation:** Protective stable oxide film on metal surface
- **Galvanic series:** Activity order of metals; guides corrosion prediction
- **Salt bridge:** Electrolytic connection maintaining charge neutrality
- **Sacrificial anode:** Active metal protecting cathode by corroding itself
- **Fuel cell:** Electrochemical device converting fuel energy to electricity continuously

⚡ QUICK RECALL TABLE

Concept	Keyword
Anode (Galvanic)	Oxidation, −ve
Cathode (Galvanic)	Reduction, +ve
Anode (Electrolytic)	Oxidation, +ve
PBR < 1	Porous, non-protective
PBR 1–2	Compact, protective
PBR > 2	Stress, spalling
Low O_2 zone	Anode → corrodes (DA)
Active metal	Anode → corrodes (GC)
Galvanizing	Zn on Fe, sacrificial
Li-ion anode	Graphite (Li intercalated)

🏆 LAST-MINUTE MEMORY ANCHORS

OXIDATION

Anode · Loss of e^- · OIL

OIL RIG: Oxidation Is Loss, Reduction Is Gain

NERNST TIP

$0.0592/n \times \log Q$

Q has products on top, reactants below

CORROSION

Less O_2 → Anode → Corrodes

Active metal → Anode in galvanic pair